



Vietnam National University HCMC
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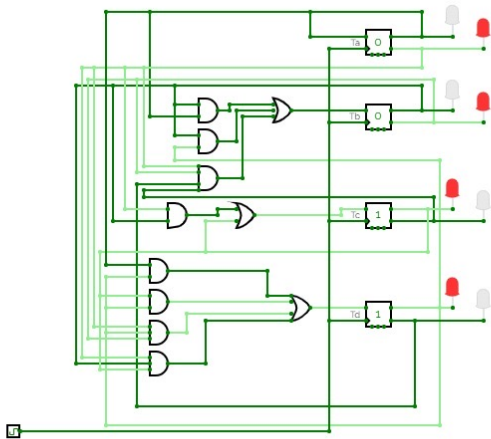


School of
Electrical Engineering

EE053IU

Digital Logic Design

Lecture 3: Logic Gates



INSTRUCTOR: Dr. Vuong Quoc Bao

Chapter Objectives

- Describe the operation of the inverter, the AND gate, and the OR gate
- Describe the operation of the NAND gate and the NOR gate
- Express the operation of NOT, AND, OR, NAND, and NOR gates with Boolean algebra
- Describe the operation of the exclusive-OR and exclusive-NOR gates
- Use logic gates in simple applications
- Recognize and use both the distinctive shape logic gate symbols and the rectangular outline logic gate symbols of ANSI/IEEE Standard 91-1984/Std. 91a-1991
- Construct timing diagrams showing the proper time relationships of inputs and outputs for the various logic gates

Chapter Objectives

- Discuss the basic concepts of programmable logic
- Make basic comparisons between the major IC technologies—CMOS and bipolar (TTL)
- Explain how the different series within the CMOS and bipolar (TTL) families differ from each other
- Define propagation delay time, power dissipation, speed-power product, and fan-out in relation to logic gates
- List specific fixed-function integrated circuit devices that contain the various logic gates

Logic gates are the vocabulary behind every digital system

Sensor decision

Automation

Parity check

Telecommunications

Address decode

Embedded systems

LUT function

FPGA

The same Boolean function can appear as a symbol, table, waveform, equation, HDL statement or PLC block.

1. The Inverter

- The inverter (NOT circuit) performs the operation called inversion or complementation.
- The inverter changes one logic level to the opposite level. In terms of bits, it changes a 1 to a 0 and a 0 to a 1.

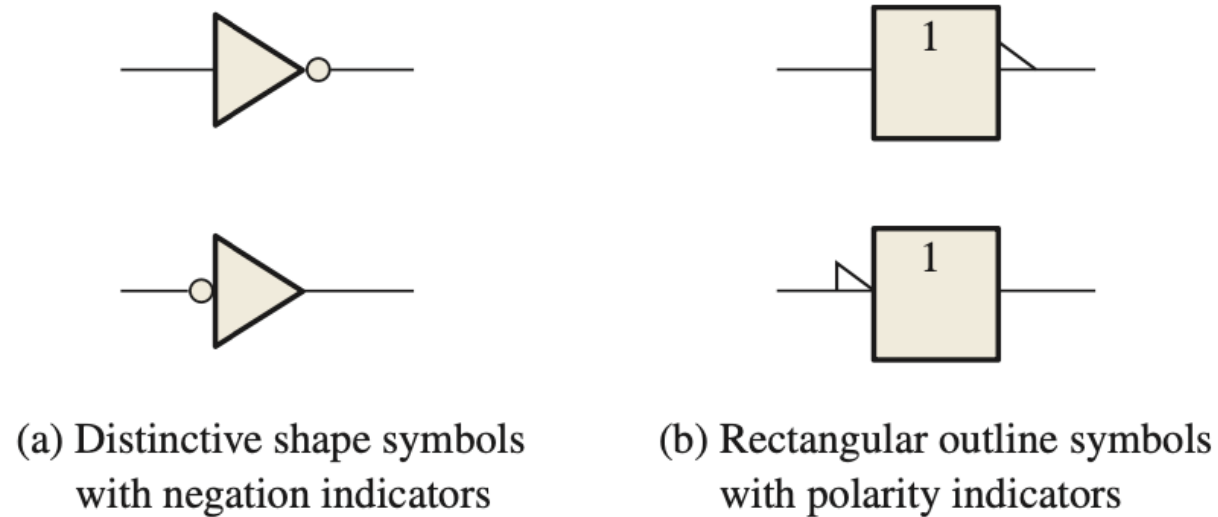


FIGURE 3-1 Standard logic symbols for the inverter (ANSI/IEEE Std. 91-1984/Std. 91a-1991).

Inverter Truth Table

- When a HIGH level is applied to an inverter input, a LOW level will appear on its output.
- When a LOW level is applied to its input, a HIGH will appear on its output.

Inverter Operation

- When the input is LOW, the output is HIGH; when the input is HIGH, the output is LOW, thereby producing an inverted output pulse.

TABLE 3-1

Inverter truth table.

Input	Output
LOW (0)	HIGH (1)
HIGH (1)	LOW (0)

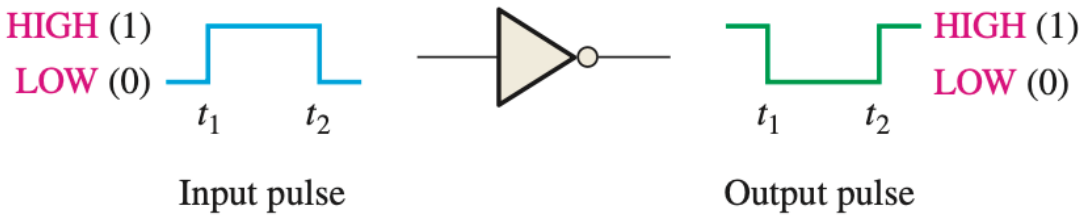


FIGURE 3-2 Inverter operation with a pulse input. Open file F03-02 to verify inverter operation. *A Multisim tutorial is available on the website.*

Timing Diagrams

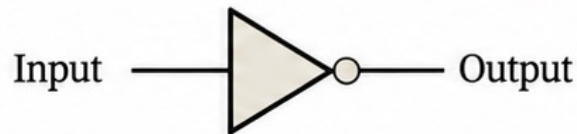
- A timing diagram shows how two or more waveforms relate in time.

EXAMPLE 3-1

A waveform is applied to an inverter in Figure 3-4. Determine the output waveform corresponding to the input and show the timing diagram. According to the placement of the bubble, what is the active output state?



FIGURE 3-4



REFERENCE

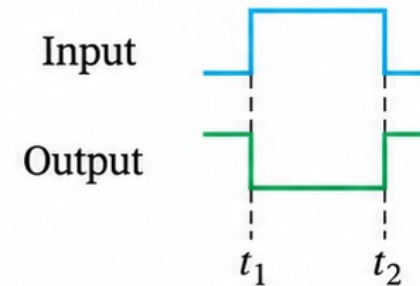


FIGURE 3-3

Timing diagram for the case in Figure 3-2.

SOLUTION

- Output is the inverse of the input.
- Active (asserted) output state: **0**.

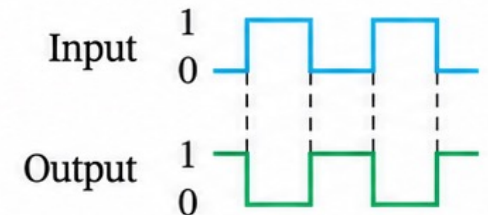


FIGURE 3-5

Logic Expression for an Inverter

- Boolean algebra uses variables and operators to describe a logic circuit.
- The operation of an inverter (NOT circuit) can be expressed as follows: If the input variable is called A and the output variable is called X, then $X = \bar{A}$

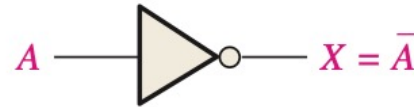


FIGURE 3-6 The inverter complements an input variable.

An Application

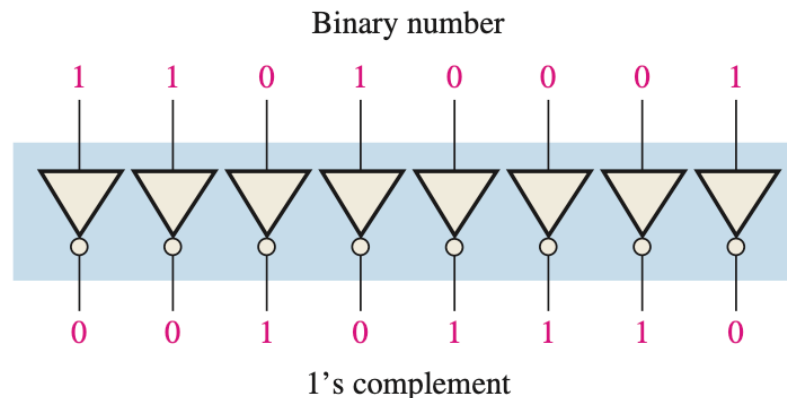
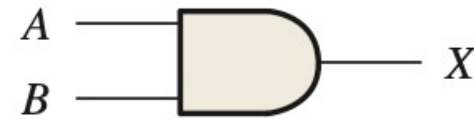


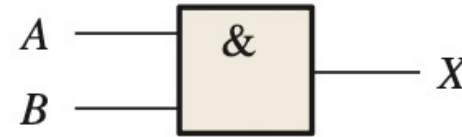
FIGURE 3-7 Example of a 1's complement circuit using inverters.

2. The AND Gate

- The AND gate is one of the basic gates that can be combined to form any logic function.
- An AND gate can have two or more inputs and performs what is known as logical multiplication.



(a) Distinctive shape



(b) Rectangular outline with the
AND (&) qualifying symbol

FIGURE 3–8 Standard logic symbols for the AND gate showing two inputs (ANSI/IEEE Std. 91-1984/Std. 91a-1991).

Operation of an AND Gate

- An AND gate produces a HIGH output only when all of the inputs are HIGH. When any of the inputs is LOW, the output is LOW.
- For a 2-input AND gate, output X is HIGH only when inputs A and B are HIGH; X is LOW when either A or B is LOW, or when both A and B are LOW.

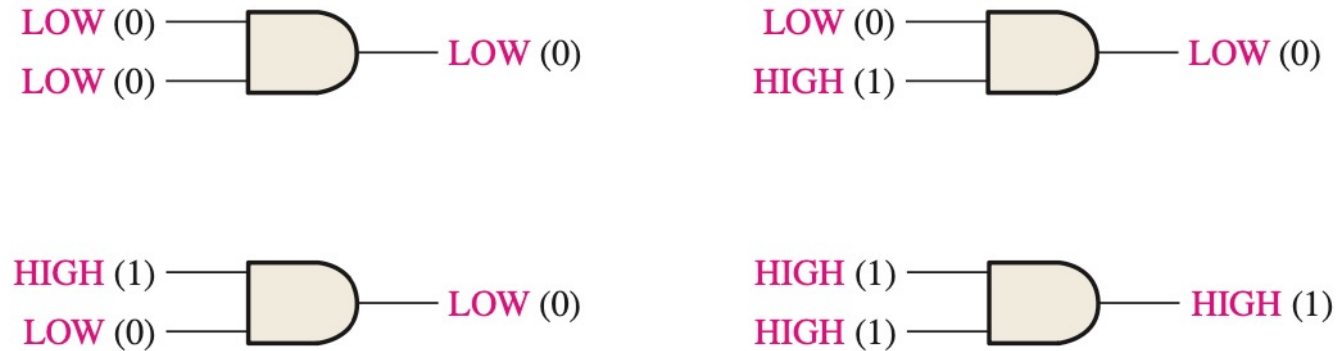


FIGURE 3-9 All possible logic levels for a 2-input AND gate. Open file F03-09 to verify AND gate operation.

AND Gate Truth Table

- The total number of possible combinations of binary inputs to a gate is determined by the following formula: $N = 2^n$
- For two input variables: $N = 2^2 = 4$ combinations
- For three input variables: $N = 2^3 = 8$ combinations
- For four input variables: $N = 2^4 = 16$ combinations

TABLE 3-2

Truth table for a 2-input AND gate.

Inputs		Output
<i>A</i>	<i>B</i>	<i>X</i>
0	0	0
0	1	0
1	0	0
1	1	1

1 = HIGH, 0 = LOW

AND Gate Truth Table

TABLE 3–3

Inputs			Output
<i>A</i>	<i>B</i>	<i>C</i>	<i>X</i>
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1

EXAMPLE 3–2

- (a) Develop the truth table for a 3-input AND gate.
- (b) Determine the total number of possible input combinations for a 4-input AND gate.

SOLUTION

- (a) There are eight possible input combinations ($2^3 = 8$) for a 3-input AND gate. The input side of the truth table (Table 3–3) shows all eight combinations of three bits. The output side is all 0s except when all three input bits are 1s.
- (b) $N = 2^4 = 16$. There are **16** possible combinations of input bits for a 4-input AND gate.

RELATED PROBLEM

Develop the truth table for a 4-input AND gate.

AND Gate Operation with Waveform Inputs

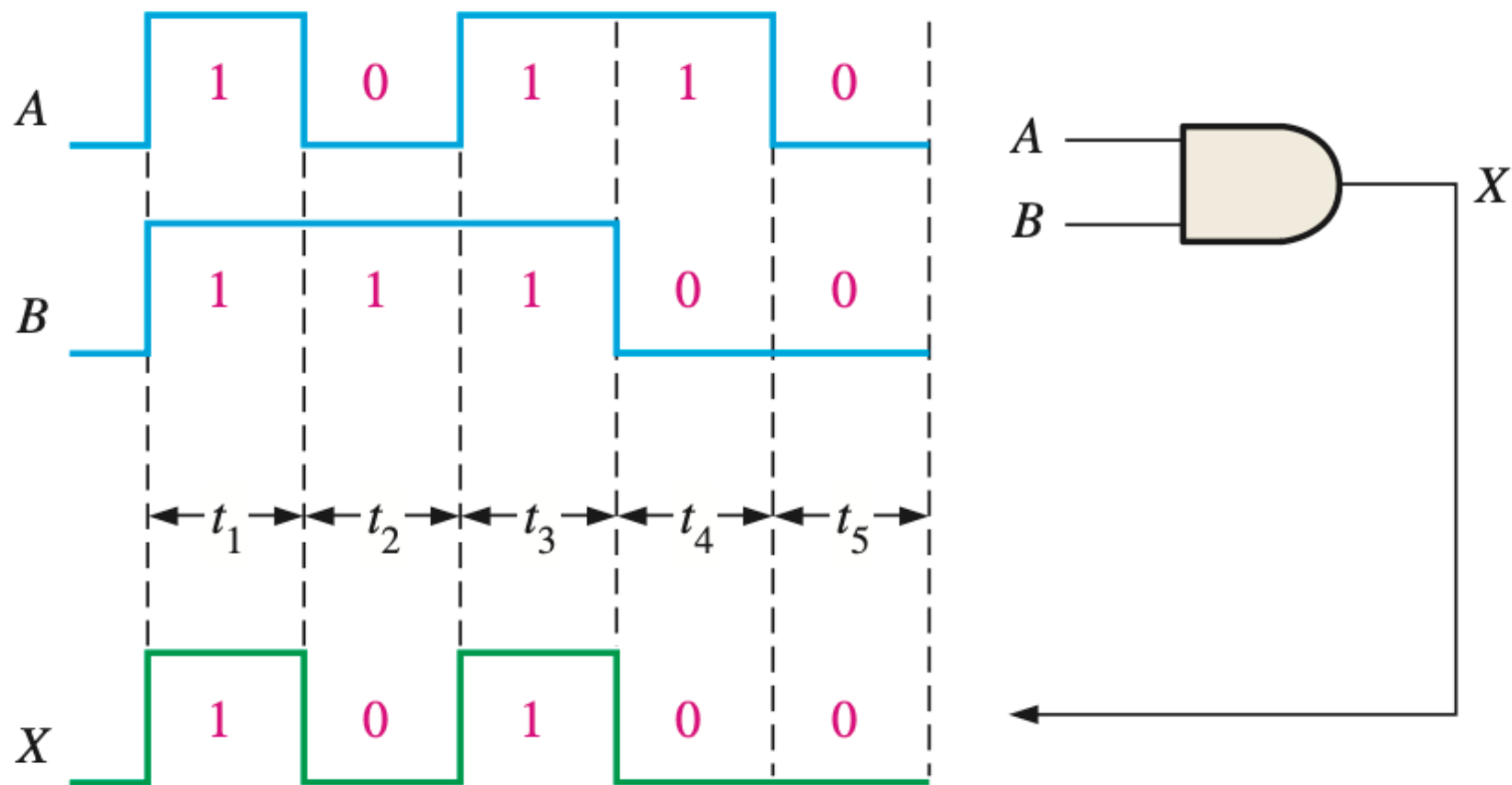
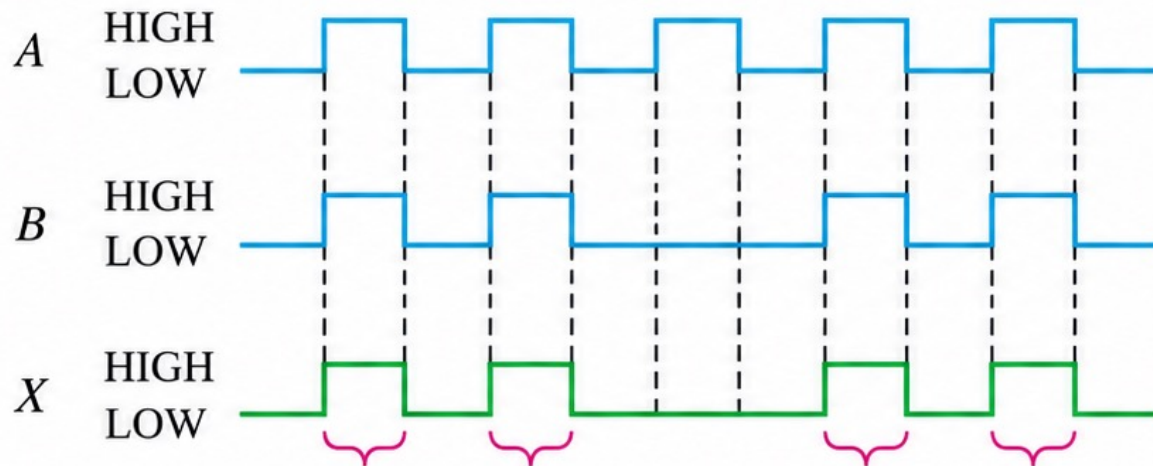


FIGURE 3-10 Example of AND gate operation with a timing diagram showing input and output relationships.

Example 3–3: AND Gate Timing Diagram

If two waveforms, *A* and *B*, are applied to the AND gate inputs as in Figure 3–11, what is the resulting output waveform?



A and *B* are both HIGH during these four time intervals; therefore, *X* is HIGH.

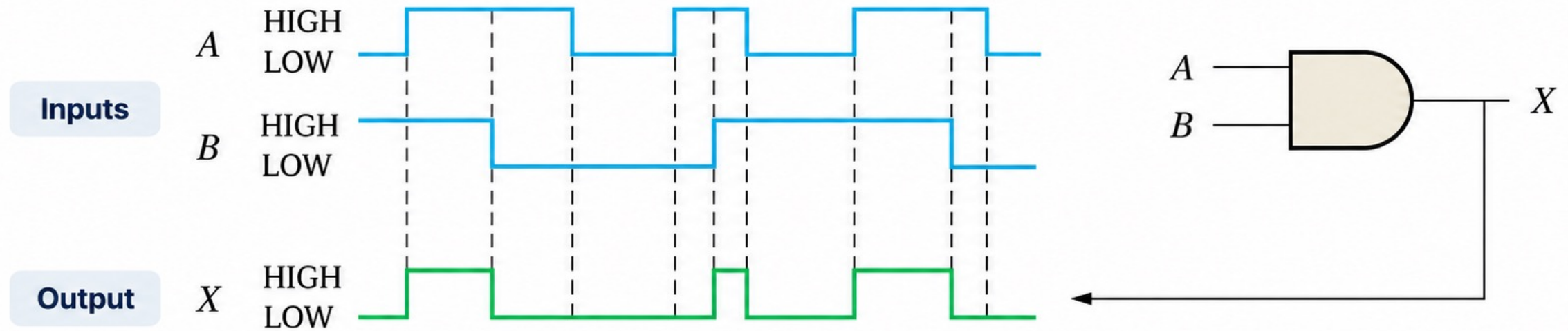


Solution

The output waveform *X* is HIGH only when both *A* and *B* are HIGH (as shown in the timing diagram).

Example 3-4: AND Gate Timing Diagram

For the two input waveforms, *A* and *B*, in Figure 3-12, show the output waveform with its proper relation to the inputs.

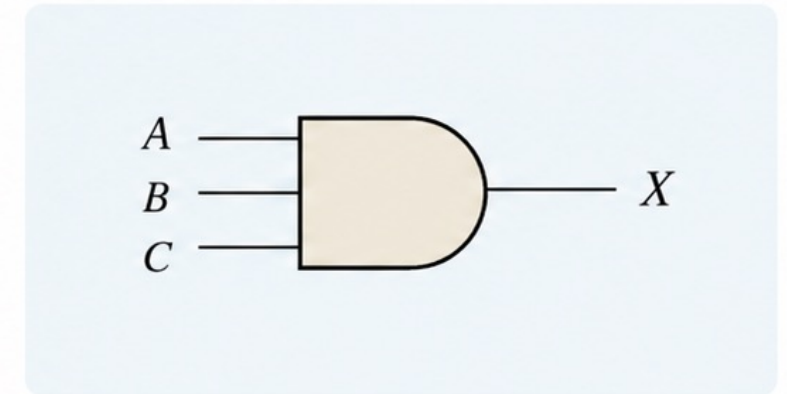
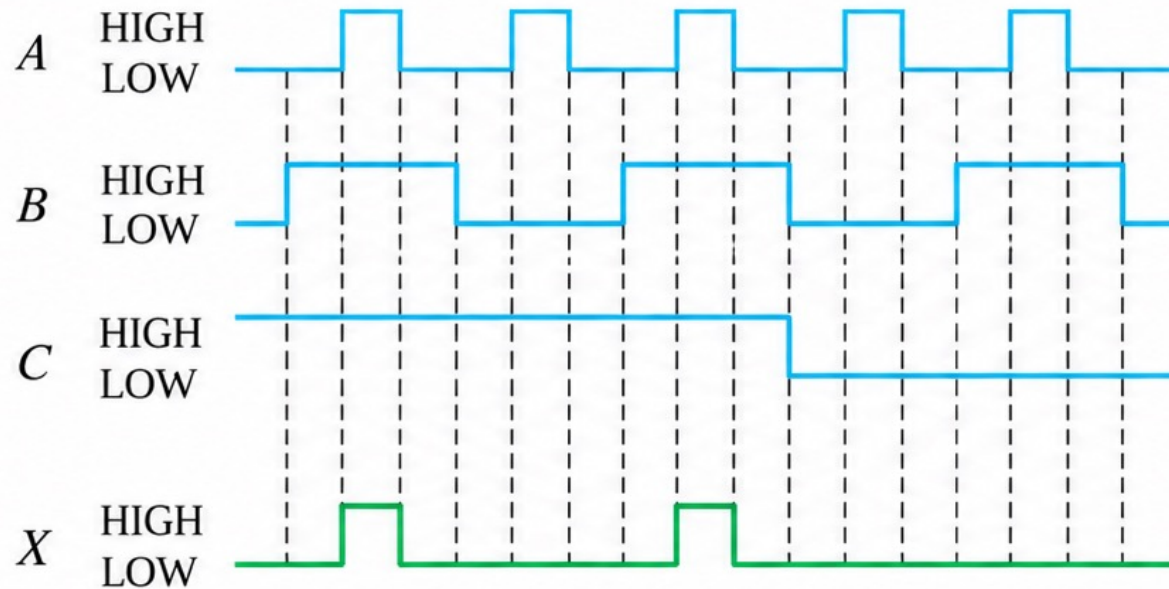


Solution

The output waveform is HIGH only when both inputs are HIGH (as shown in the timing diagram).

Example 3-5: 3-Input AND Gate Timing Diagram

For the 3-input AND gate in Figure 3-13, determine the output waveform in relation to the inputs.



Solution

The output waveform X is HIGH only when all three inputs A , B , and C are HIGH.

Logic Expressions for an AND Gate

- Boolean multiplication follows the same basic rules governing binary multiplication.

0 • 0 = 0

0 • 1 = 0

1 • 0 = 0

1 • 1 = 1

- Boolean multiplication is the same as the AND function.

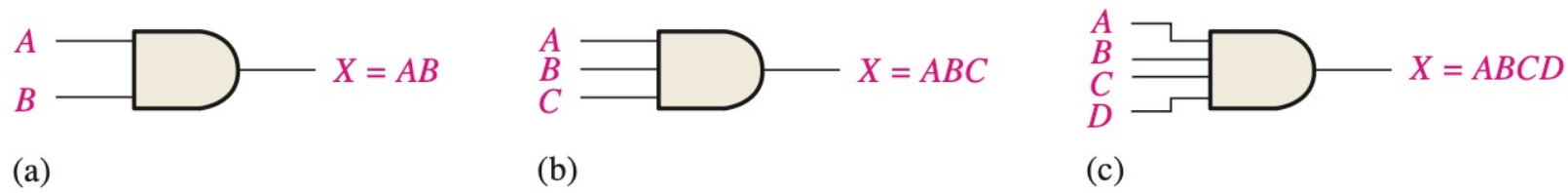


FIGURE 3-15 Boolean expressions for AND gates with two, three, and four inputs.

TABLE 3-4

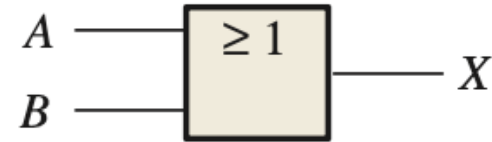
A	B	AB = X
0	0	0 • 0 = 0
0	1	0 • 1 = 0
1	0	1 • 0 = 0
1	1	1 • 1 = 1

3. The OR Gate

- The OR gate is another of the basic gates from which all logic functions are constructed.
- An OR gate can have two or more inputs and performs what is known as logical addition.



(a) Distinctive shape



(b) Rectangular outline with the
OR (≥ 1) qualifying symbol

FIGURE 3-18 Standard logic symbols for the OR gate showing two inputs (ANSI/IEEE Std. 91-1984/Std. 91a-1991).

Operation of an OR Gate

- An OR gate produces a HIGH on the output when any of the inputs is HIGH. The output is LOW only when all of the inputs are LOW.
- For a 2-input OR gate, output X is HIGH when either input A or input B is HIGH, or when both A and B are HIGH; X is LOW only when both A and B are LOW.

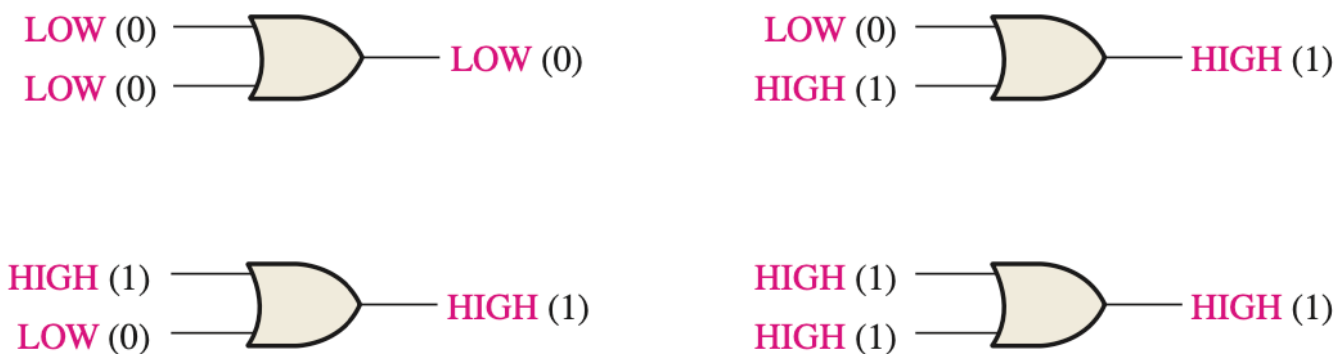


FIGURE 3-19 All possible logic levels for a 2-input OR gate. Open file F03-19 to verify OR gate operation.

TABLE 3-5
Truth table for a 2-input OR gate.

Inputs		Output
A	B	X
0	0	0
0	1	1
1	0	1
1	1	1

1 = HIGH, 0 = LOW

OR Gate Operation with Waveform Inputs

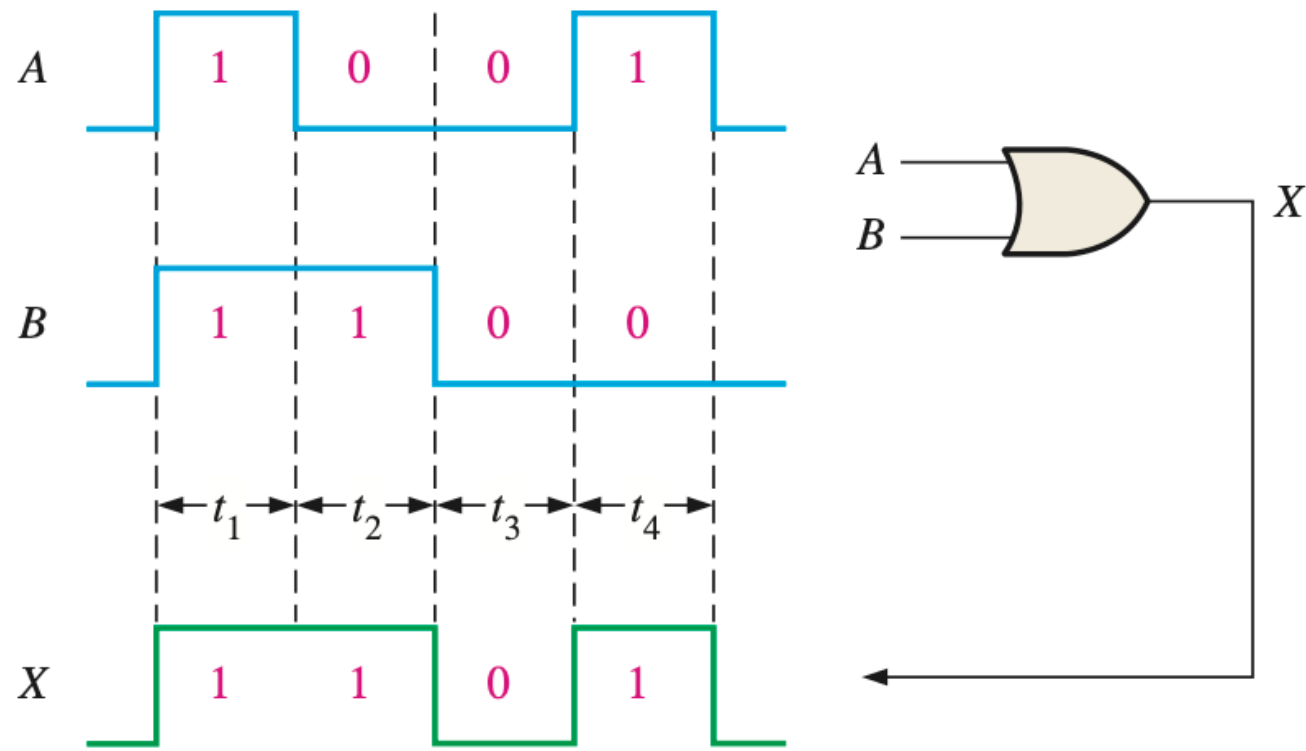
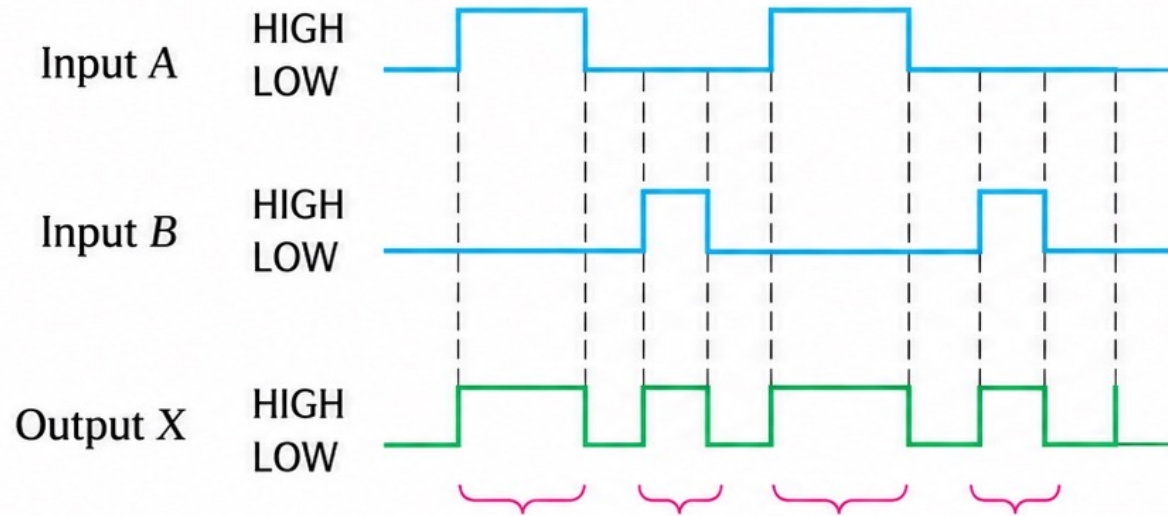


FIGURE 3–20 Example of OR gate operation with a timing diagram showing input and output time relationships.

Example 3-7 OR Gate Timing Diagram

If the input waveforms *A* and *B* in Figure 3-21 are applied to the OR gate, what is the output waveform?



When either input or both inputs are HIGH, the output is HIGH.

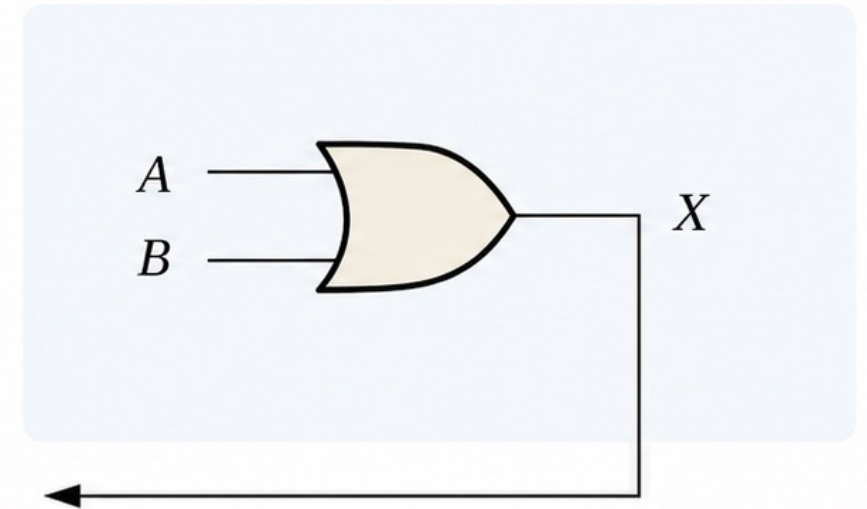


FIGURE 3-21

Solution

The output waveform *X* is HIGH when either or both inputs are HIGH (as shown in the timing diagram).

Example 3–8 OR Gate Timing Diagram

For the two input waveforms, *A* and *B*, in Figure 3–22, show the output waveform with its proper relation to the inputs.

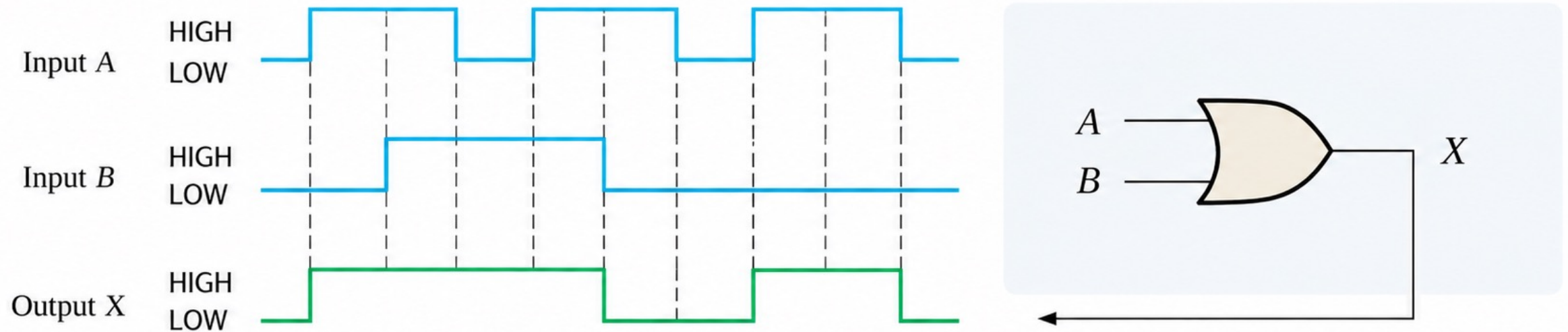


FIGURE 3–22

Solution

The output waveform *X* is HIGH when either or both inputs are HIGH (as shown in the timing diagram).

Example 3–9 3-Input OR Gate Timing Diagram

For the 3-input OR gate in Figure 3–23, determine the output waveform in proper time relation to the inputs.

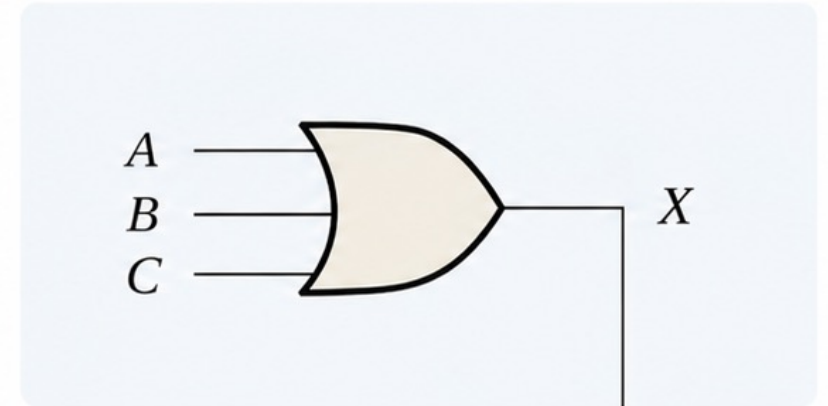
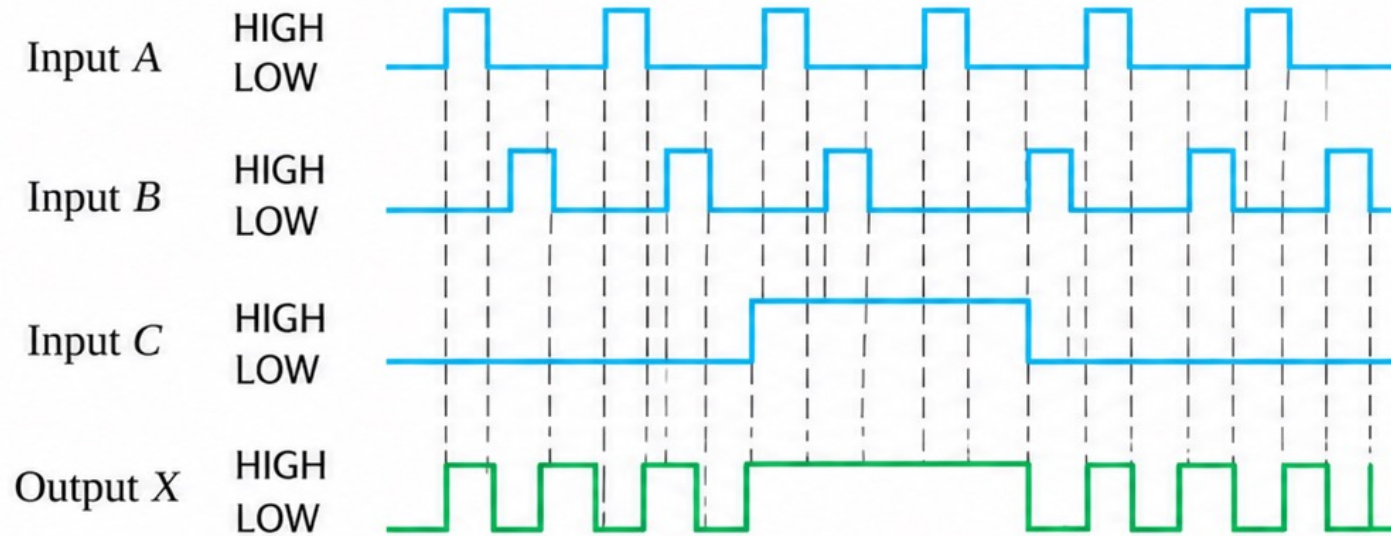


FIGURE 3–23

Solution

The output X is HIGH when one or more inputs are HIGH (as shown in the timing diagram).

Related Problem

Determine the output waveform if input C is always LOW.

Logic Expressions for an OR Gate

- The logical OR function of two variables is represented mathematically by a + between the two variables.

$0 + 0 = 0$
$0 + 1 = 1$
$1 + 0 = 1$
$1 + 1 = 1$

- Boolean multiplication is the same as the OR function.

TABLE 3-6		
<i>A</i>	<i>B</i>	$A + B = X$
0	0	$0 + 0 = 0$
0	1	$0 + 1 = 1$
1	0	$1 + 0 = 1$
1	1	$1 + 1 = 1$

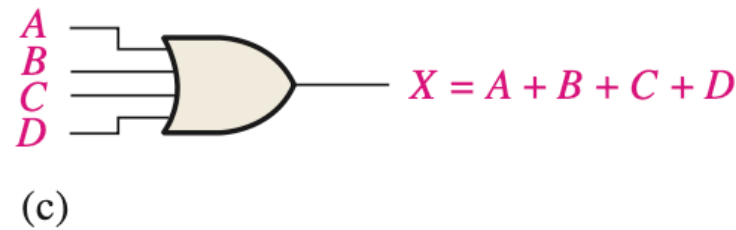
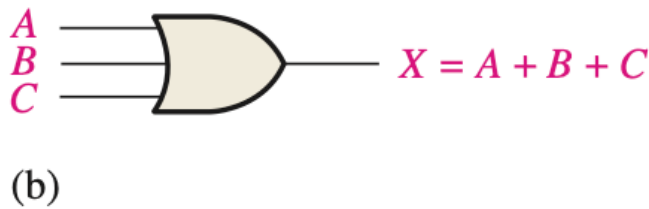
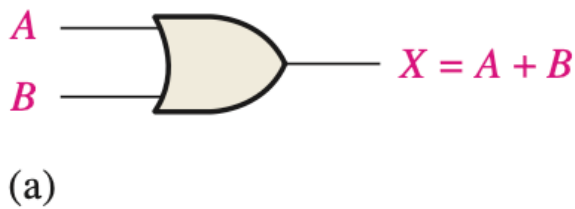
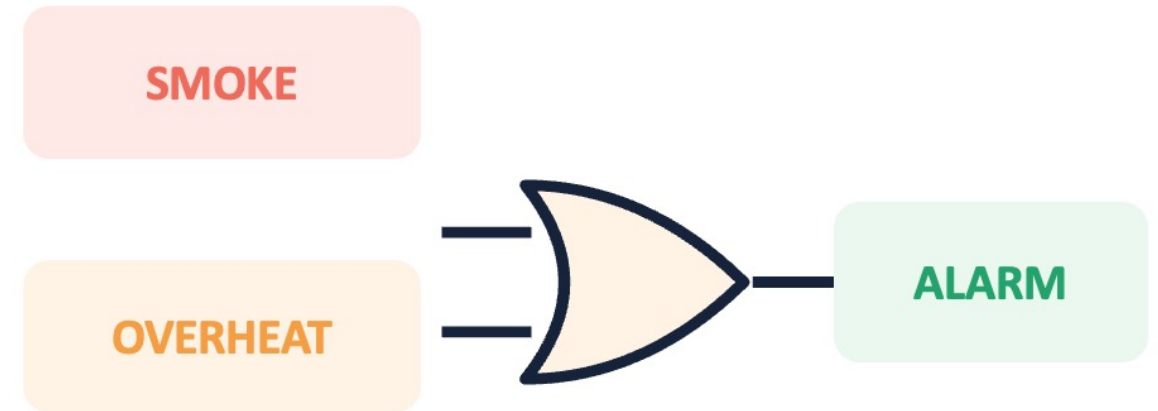
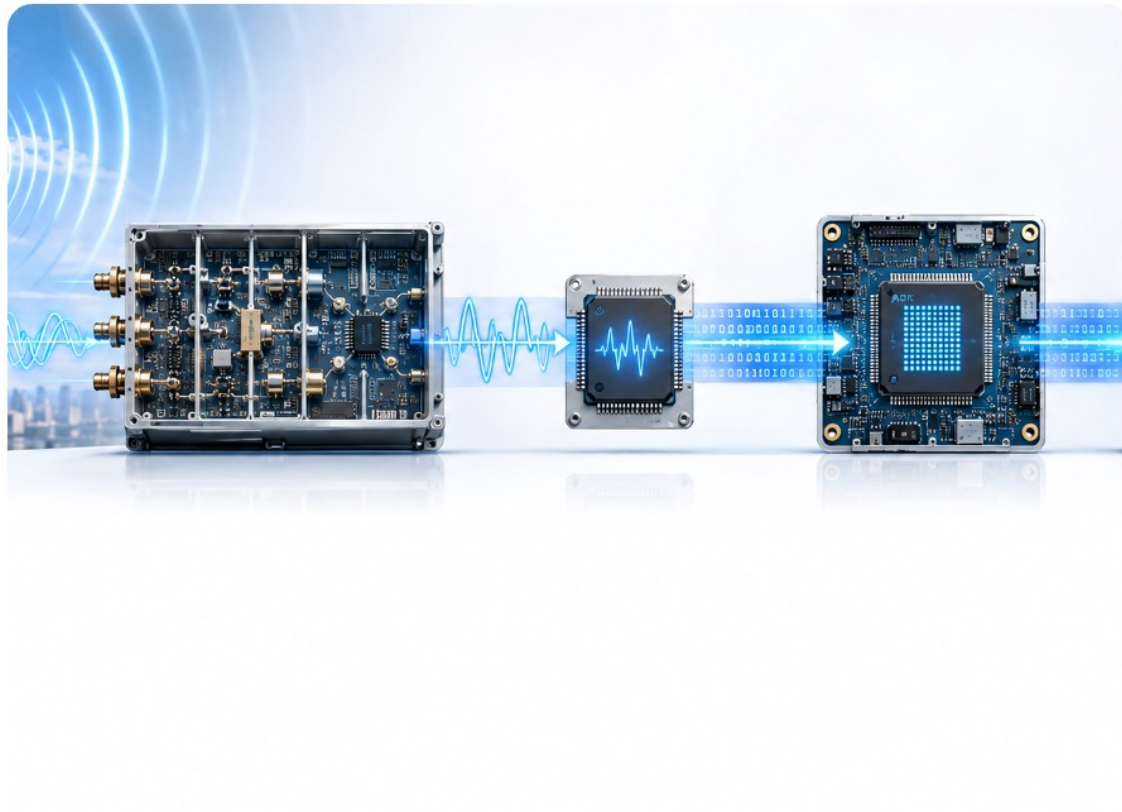


FIGURE 3-24 Boolean expressions for OR gates with two, three, and four inputs.

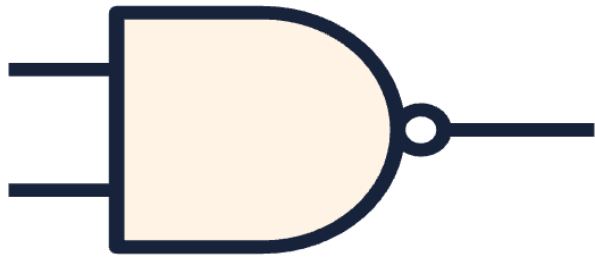
Telecom and automation use OR to aggregate alarms



$$\text{ALARM} = \text{SMOKE} + \text{OVERHEAT}$$

4. The NAND Gate

A NAND gate is LOW only when every input is HIGH



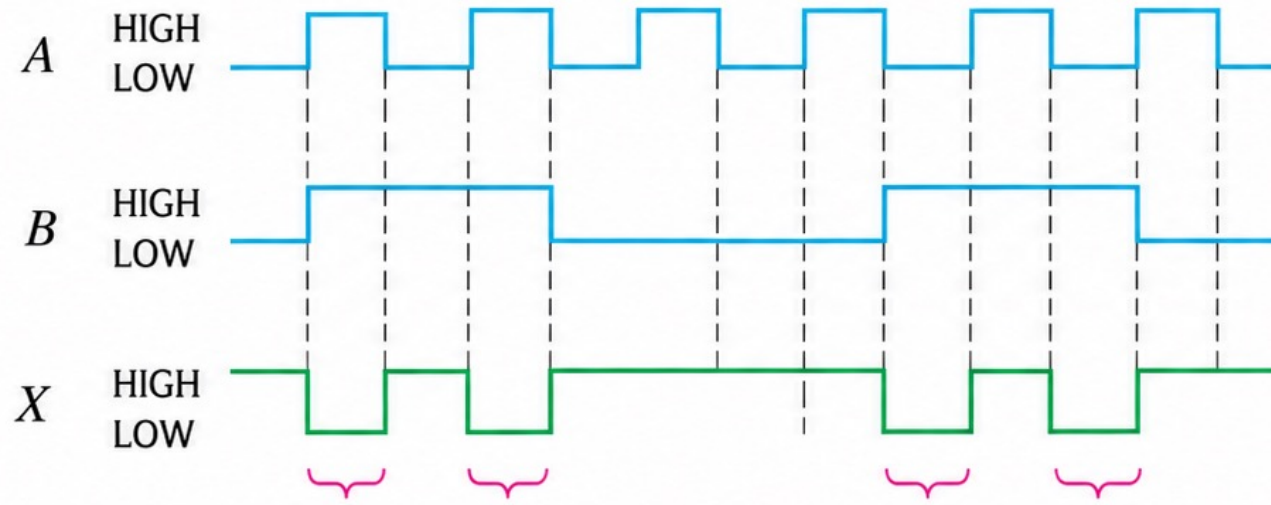
A	B	X
0	0	1
0	1	1
1	0	1
1	1	0

$$X = \overline{A \cdot B}$$

Correct symbol = AND shape + output bubble

Example 3-10 NAND Gate Timing Diagram

If the two waveforms *A* and *B* in Figure 3-28 are applied to the NAND gate inputs, determine the output waveform.



A and *B* are both HIGH during these four time intervals; therefore, *X* is LOW.

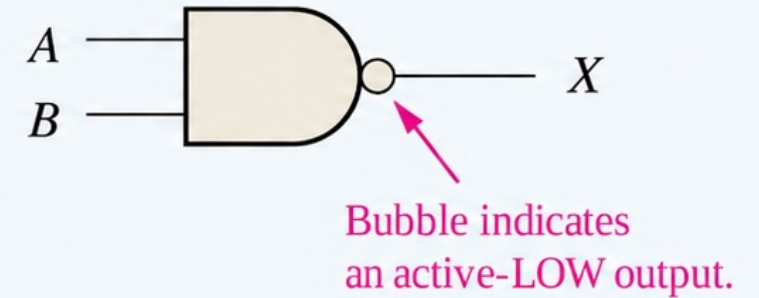


FIGURE 3-28

Solution

The output waveform *X* is LOW only when both inputs *A* and *B* are HIGH (as shown in the timing diagram).

Example 3-11 3-Input NAND Gate Timing Diagram

Show the output waveform for the 3-input NAND gate in Figure 3-29 with its proper time relationship to the inputs.

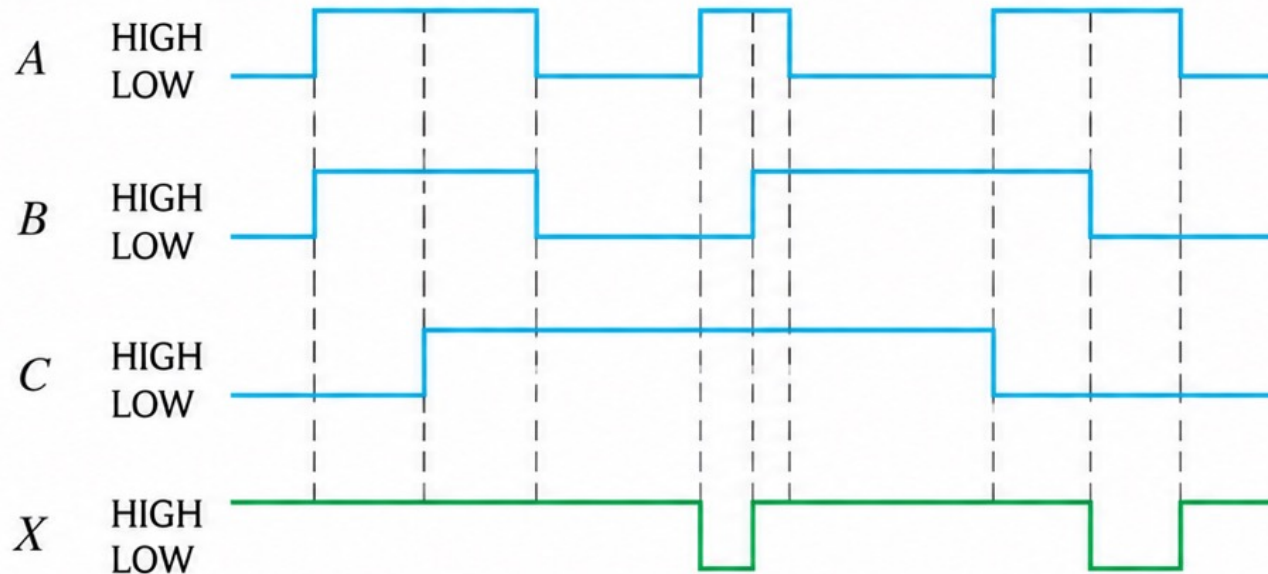
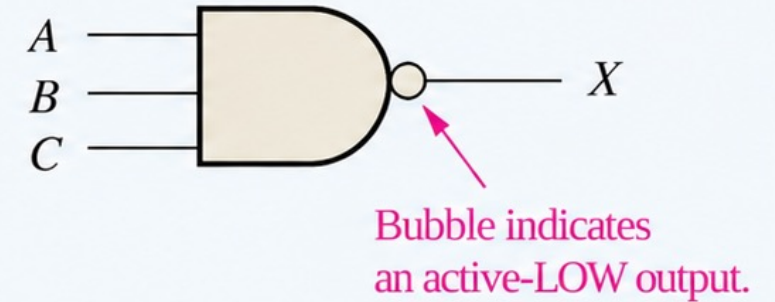


FIGURE 3-29



Solution

The output waveform X is LOW only when all three inputs A , B , and C are HIGH (as shown in the timing diagram).

Negative-OR

Equivalent Operation of a NAND Gate

- 1 A NAND gate outputs **HIGH** when one or more inputs are **LOW** → called **negative-OR**.
- 2 For a 2-input NAND gate, output X is **HIGH** if
 - A is **LOW**, or
 - B is **LOW**, or
 - both A and B are **LOW**.



Key Takeaway

NAND = NOT (A OR B)

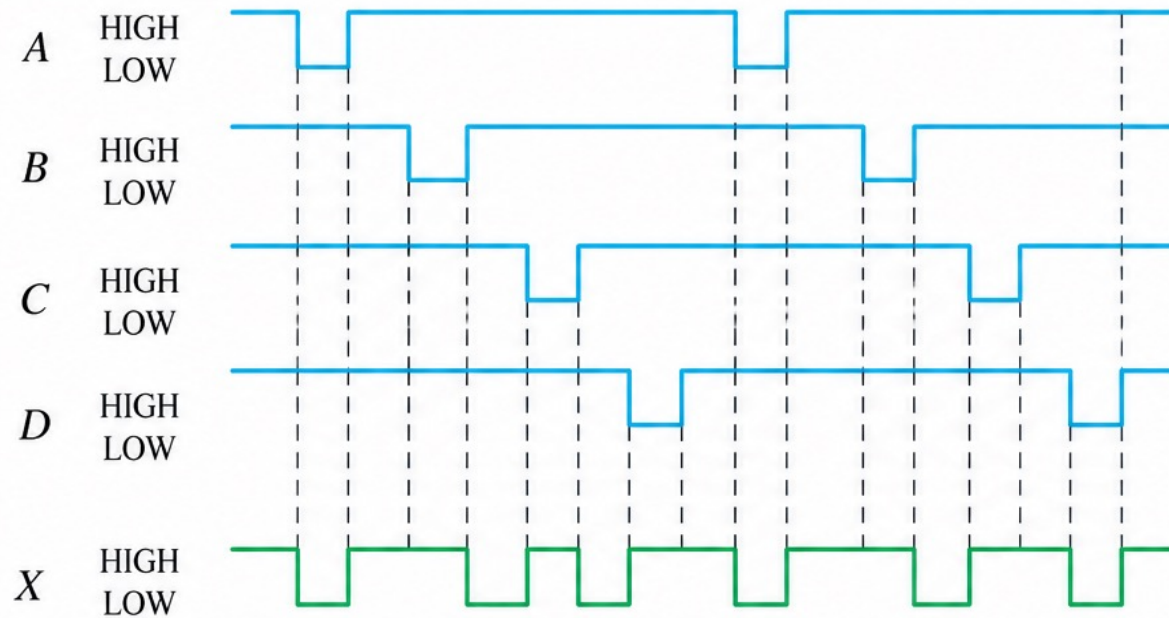
It produces a **HIGH** output when any input is **LOW**.



FIGURE 3–30 ANSI/IEEE standard symbols for the two equivalent operations of a NAND gate.

Example 3–14 4-Input NAND Gate (Negative-OR)

For the 4-input NAND gate in Figure 3–33, operating as a negative-OR gate, determine the output with respect to the inputs.



Bubbles indicate active-LOW inputs.

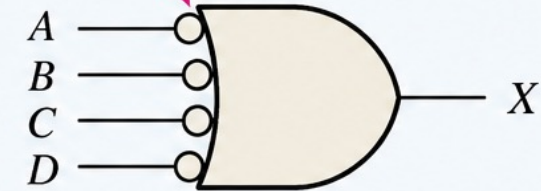


FIGURE 3–33

Solution

The output waveform X is HIGH whenever **any** input is LOW (as shown in the timing diagram).

Logic Expressions for a NAND Gate

- The Boolean expression for the output of a 2-input NAND gate is

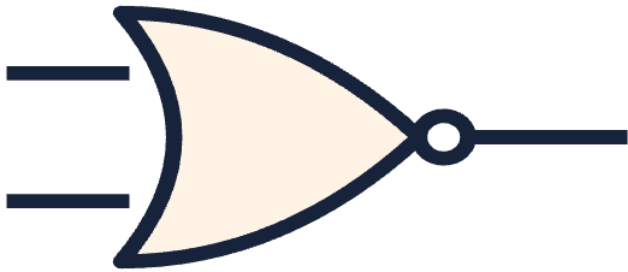
TABLE 3–8

<i>A</i>	<i>B</i>	$\overline{AB} = X$
0	0	$\overline{0 \cdot 0} = \overline{0} = 1$
0	1	$\overline{0 \cdot 1} = \overline{0} = 1$
1	0	$\overline{1 \cdot 0} = \overline{0} = 1$
1	1	$\overline{1 \cdot 1} = \overline{1} = 0$

- A bar over a variable or variables indicates an inversion.

5. The NOR Gate

A NOR gate is HIGH only when every input is LOW



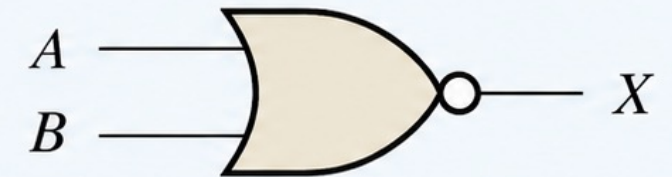
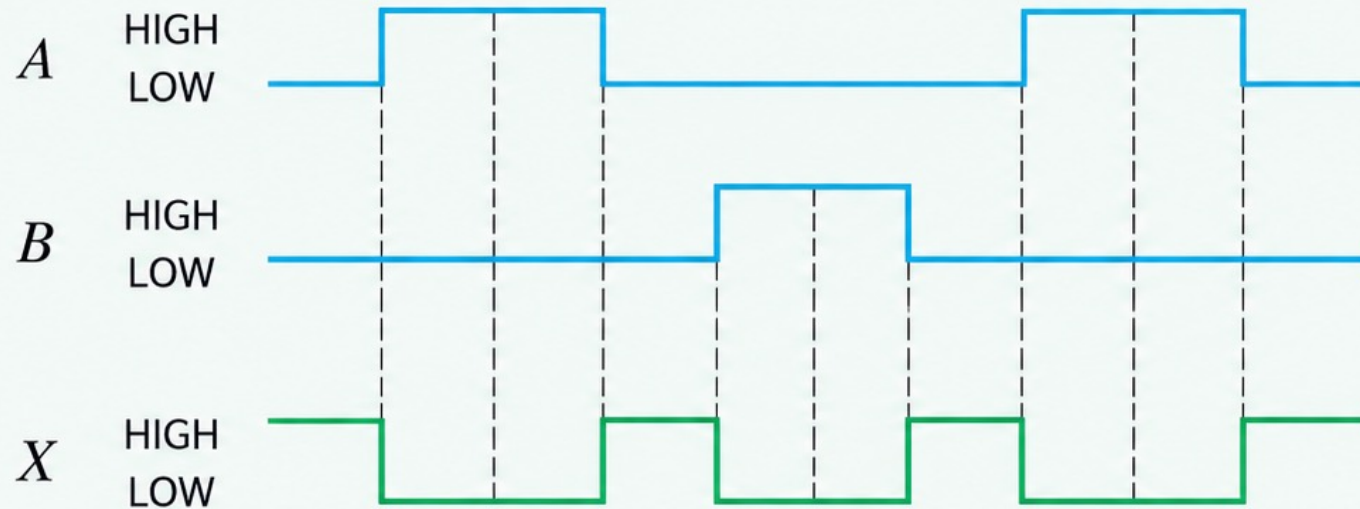
A	B	X
0	0	1
0	1	0
1	0	0
1	1	0

$$X = \overline{A + B}$$

Correct symbol = OR shape + output bubble

Example 3–15 2-Input NOR Gate Timing Diagram

If the two waveforms in Figure 3–36 are applied to a NOR gate, what is the resulting output waveform?



2-Input NOR Gate

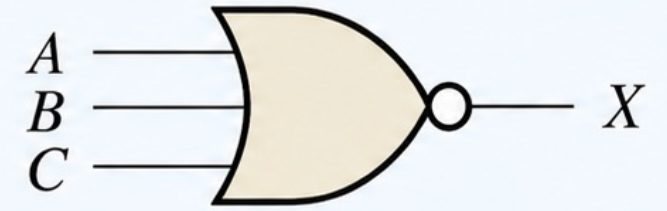
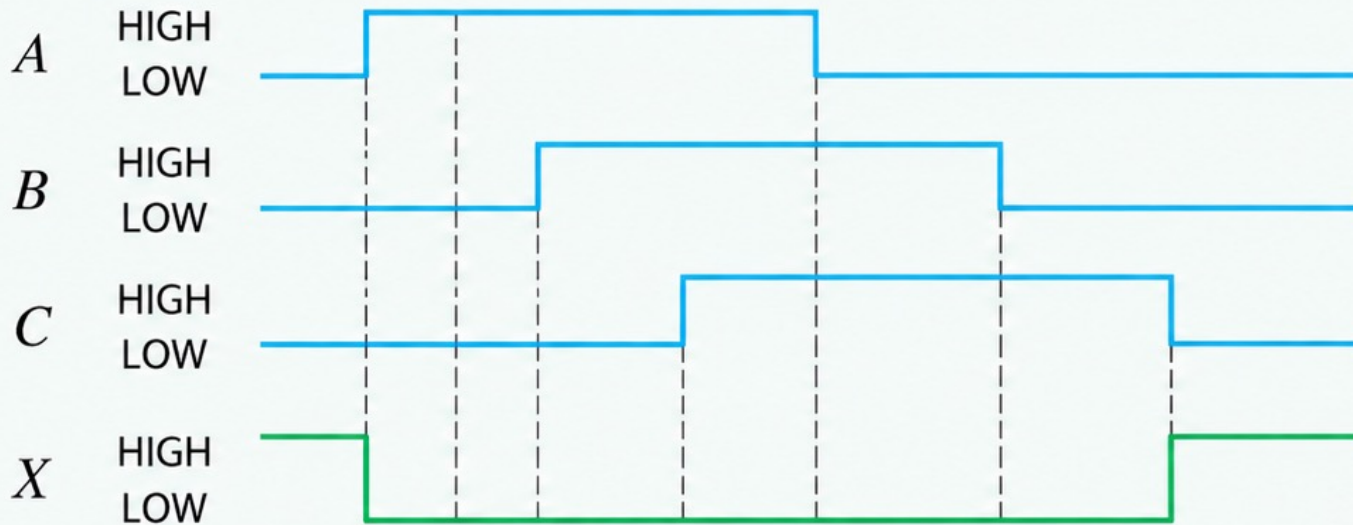
$$X = \neg(A + B)$$

Solution

The output X is LOW whenever **any** input is HIGH (as shown in the timing diagram).

Example 3–16 3-Input NOR Gate Timing Diagram

Show the output waveform for the 3-input NOR gate in Figure 3–37 with the proper time relationship to the inputs.



3-Input NOR Gate

$$X = \neg(A + B + C)$$

Solution

The output X is LOW when **any** input is HIGH (as shown in the timing diagram).

Negative-AND Equivalent Operation of a NOR Gate

1 A NOR gate has another inherent operation
→ called **negative-AND**.

2 For a 2-input NOR gate (negative-AND),
output X is **HIGH** only when both inputs
 A and B are **LOW**.



Key Takeaway

NOR = NOT (A OR B)

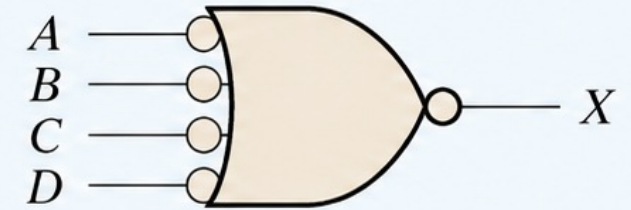
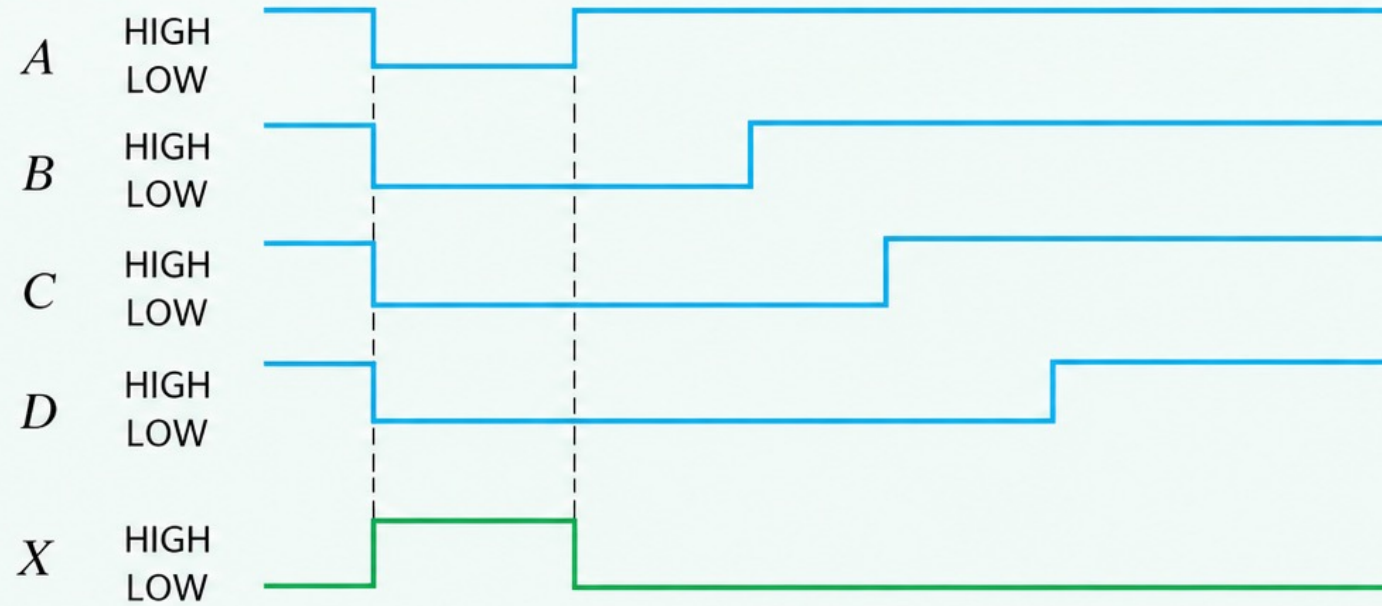
It produces a **HIGH** output
only when all inputs are **LOW**.



FIGURE 3-38 Standard symbols representing the two equivalent operations of a NOR gate.

Example 3–19 4-Input NOR Gate (Negative-AND)

For the 4-input NOR gate operating as a negative-AND in Figure 3–41, determine the output relative to the inputs.



4-Input NOR Gate

$$X = \neg(A + B + C + D)$$

Solution

The output X is **HIGH** only when **all inputs are LOW** (as shown in the timing diagram).

Logic Expressions for a NOR Gate

- The Boolean expression for the output of a 2-input NOR gate can be written as

TABLE 3–10

<i>A</i>	<i>B</i>	$\overline{A + B} = X$
0	0	$\overline{0 + 0} = \overline{0} = 1$
0	1	$\overline{0 + 1} = \overline{1} = 0$
1	0	$\overline{1 + 0} = \overline{1} = 0$
1	1	$\overline{1 + 1} = \overline{1} = 0$

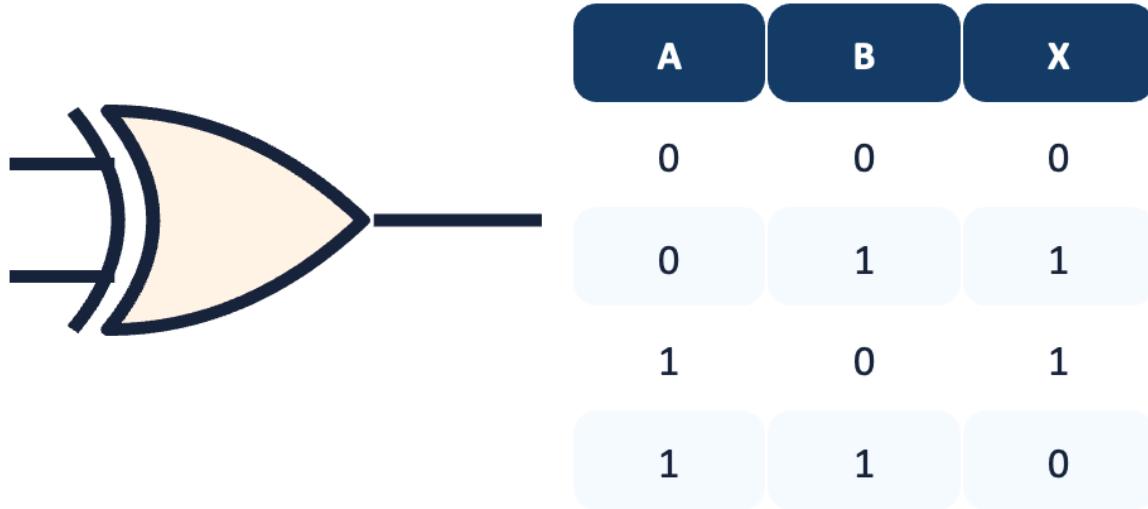
- The NOR expression can be extended to more than two input variables by including additional letters to represent the other variables.

6. The Exclusive-OR and Exclusive-NOR Gates

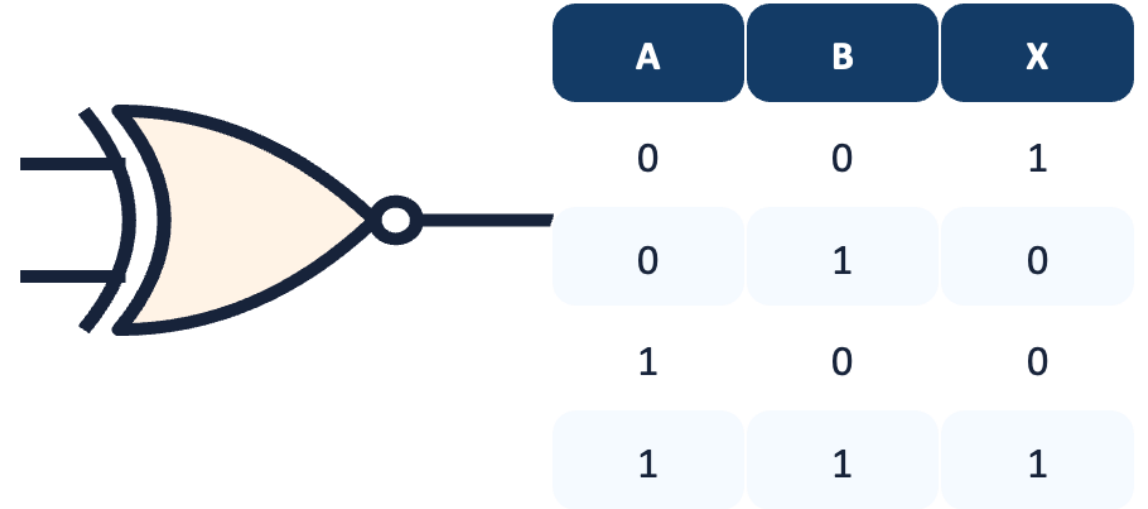
XOR detects difference; XNOR detects equality

These gates compare two logic levels and underpin parity, edge detection and comparators.

XOR is HIGH for unequal inputs; XNOR for equal inputs



$$X = A \oplus B = \overline{A}B + A\overline{B}$$



$$X = \overline{A \oplus B} = AB + \overline{A}\overline{B}$$

XOR output marks intervals where A and B differ

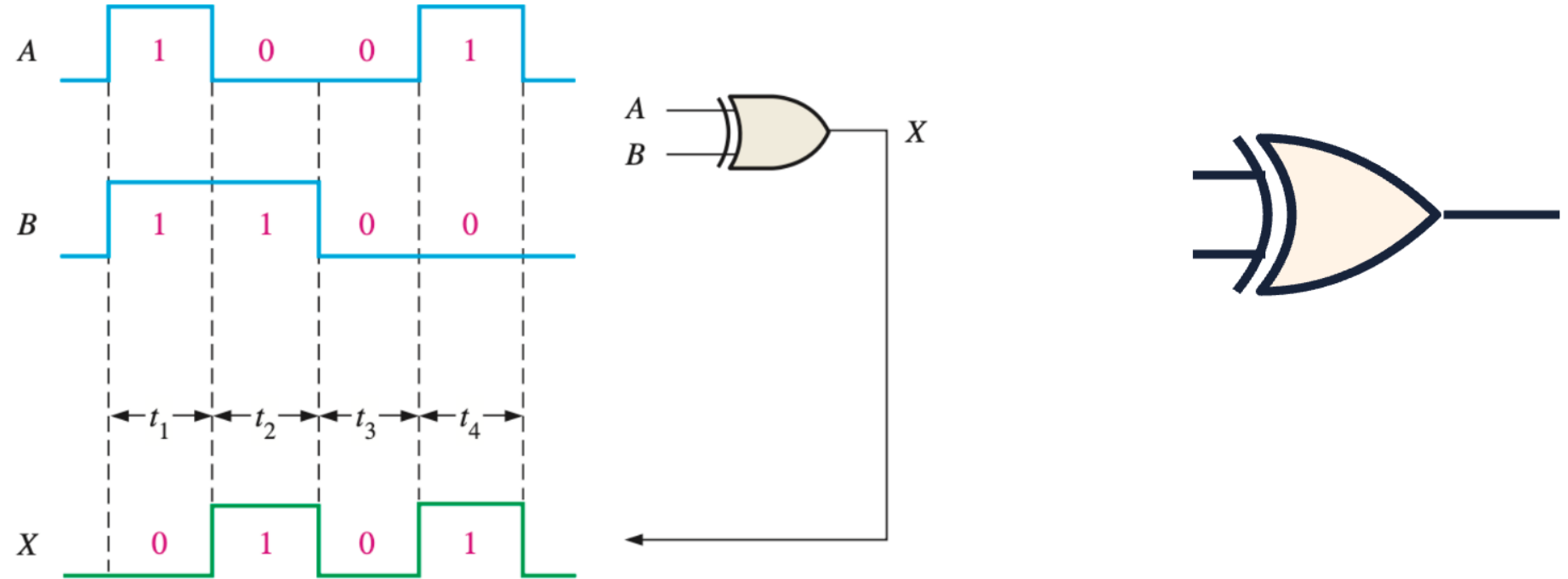
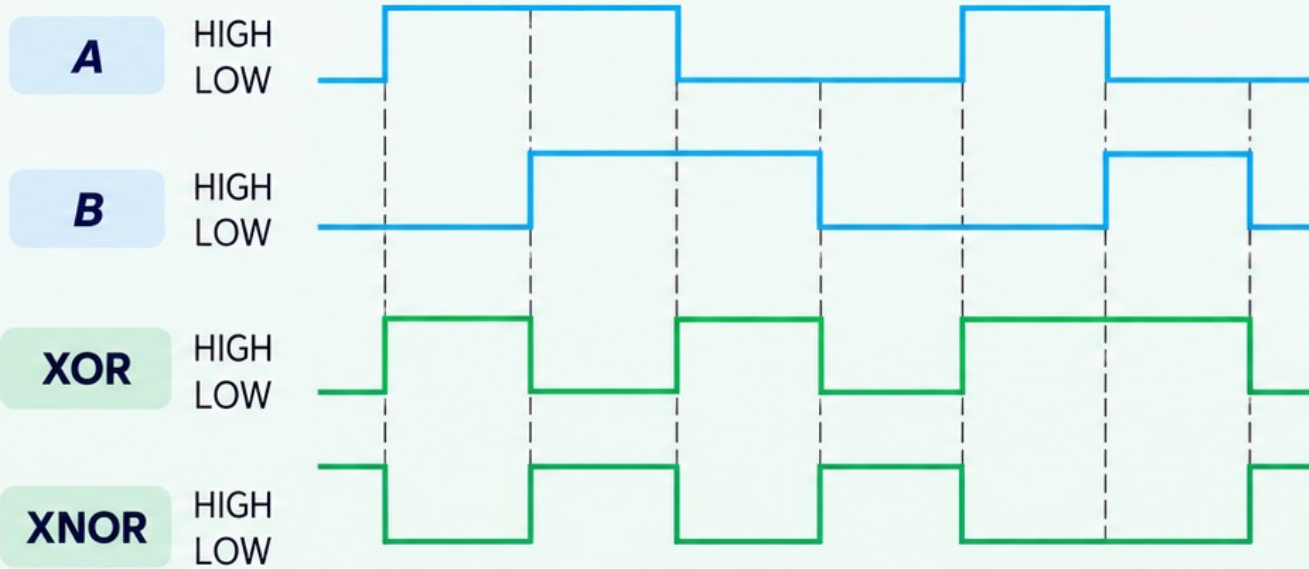


FIGURE 3–47 Example of exclusive-OR gate operation with pulse waveform inputs.

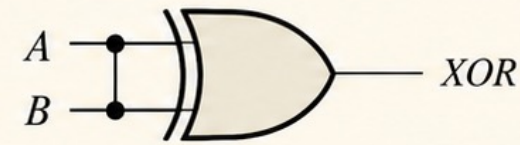
XNOR is the exact complement of this output

Example 3–21 XOR and XNOR Output Waveforms

Determine the output waveforms for the XOR gate and for the XNOR gate, given the input waveforms *A* and *B*.



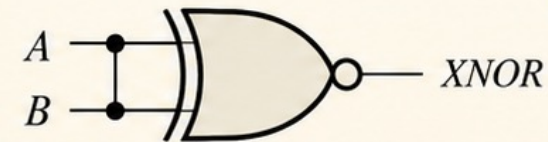
XOR Gate



$$\text{XOR} = A \oplus B$$

HIGH when inputs are **different**

XNOR Gate



$$\text{XNOR} = \overline{A \oplus B}$$

HIGH when inputs are **the same**



Solution

- XOR output is **HIGH** only when the inputs are **different**.
- XNOR output is **HIGH** only when the inputs are **the same**.

The AND Array

A matrix of AND gates with programmable interconnections (used in most PLDs)

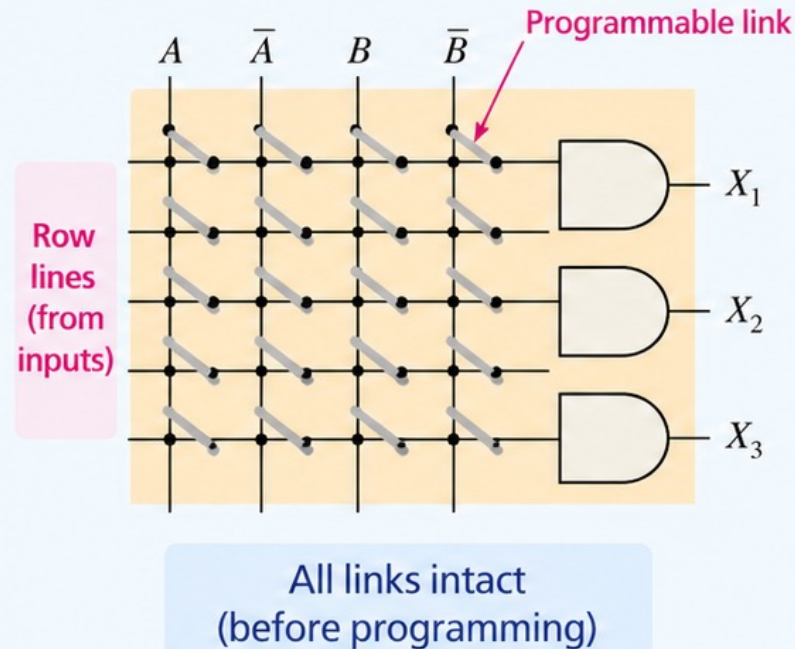
1 Most PLDs use an **AND array**.

2 It consists of AND gates and a matrix of interconnections with a **programmable link** at each cross point.

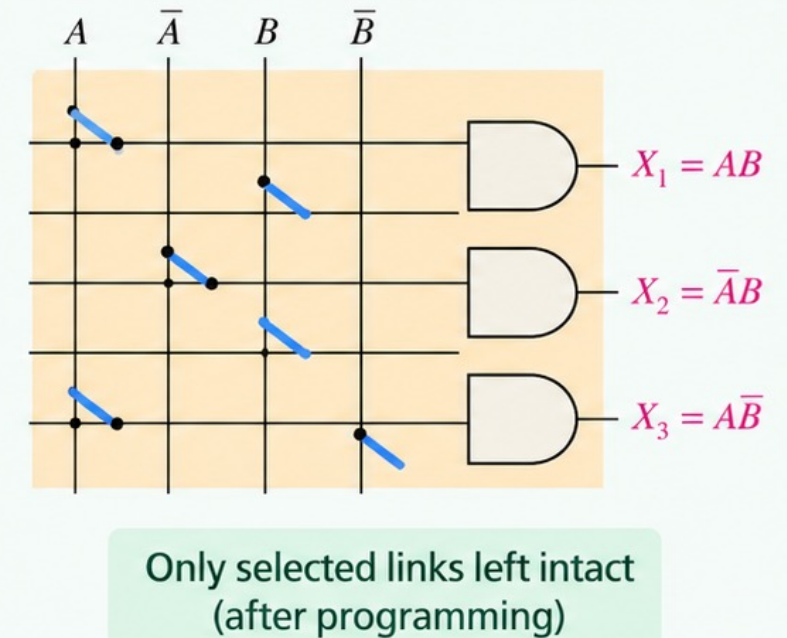
3 Each link can be **opened** (disconnected) or **left intact** (connected).

4 For each input to an AND gate, **only one link** is left intact to connect the desired variable to the gate input.

(a) Unprogrammed AND array



(b) Programmed AND array



Key Idea

The AND array uses **programmable links** to connect input variables to AND gates. For each gate input, **only one link** is left intact.

Example 3–22 Programming an AND Array

Show the AND array in Figure 3–49(a) programmed for the following outputs: $X_1 = A\bar{B}$, $X_2 = \bar{A}B$, and $X_3 = \bar{A}\bar{B}$.

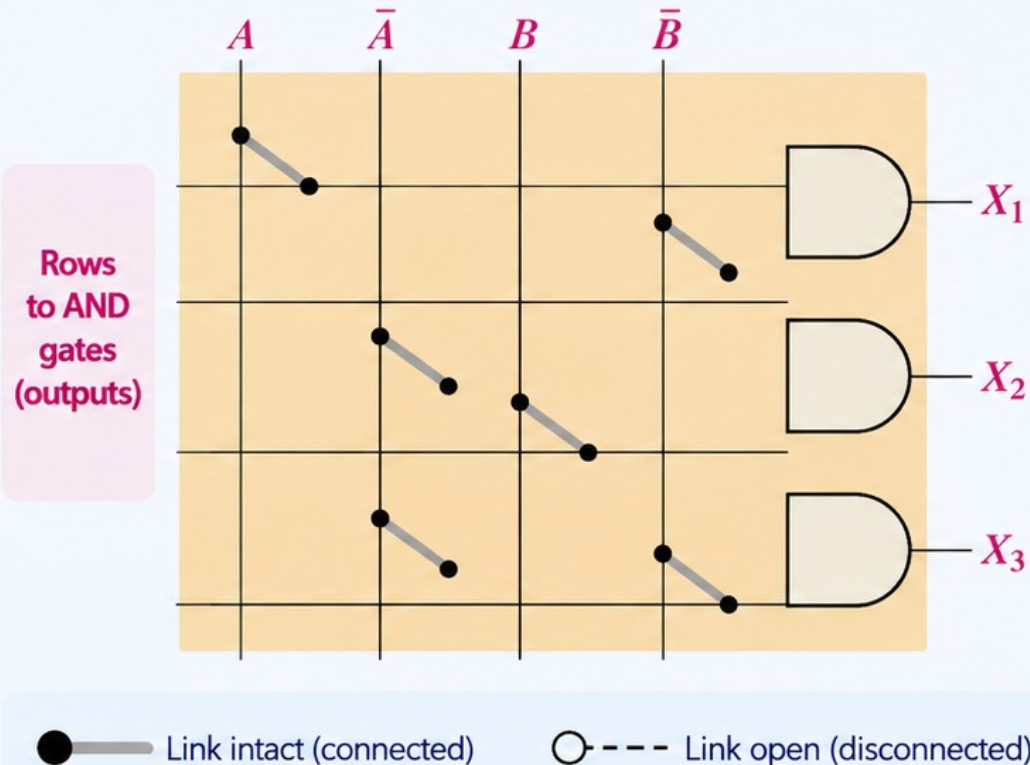
Steps

- 1 For each output, select the input variables needed for the AND term.
- 2 Leave the corresponding programmable links **intact** (connected).
- 3 All other links remain **open** (disconnected).

Outputs

$$\begin{aligned} X_1 &= A\bar{B} && (A \text{ and } \bar{B}) \\ X_2 &= \bar{A}B && (\bar{A} \text{ and } B) \\ X_3 &= \bar{A}\bar{B} && (\bar{A} \text{ and } \bar{B}) \end{aligned}$$

Programmed AND array (Figure 3–50)



Result

- 1 $X_1 = A\bar{B}$
Connect **A** and **$\bar{B}$**
- 2 $X_2 = \bar{A}B$
Connect **$\bar{A}$** and **B**
- 3 $X_3 = \bar{A}\bar{B}$
Connect **$\bar{A}$** and **$\bar{B}$**



Key Idea

Each AND gate output is formed by leaving only the **required links intact**.



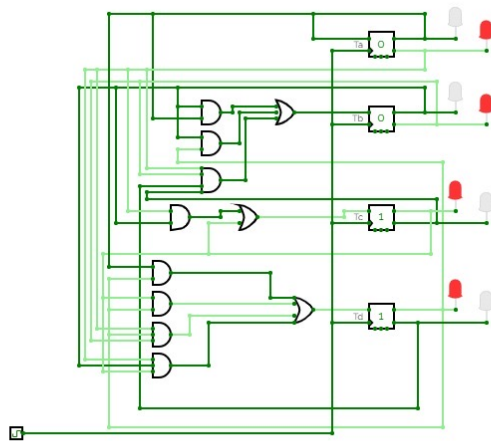
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THE END

Lecture 3: Logic Gates



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